Carbon Emission Scenarios: Switching from Ttareungyi (Public Bike-Sharing) to Passenger Cars

1. Passenger-car emission factor

Passenger-car emission factors can vary by fuel efficiency and power source. According to the *Guidelines for Applying Emission Reduction Units by Local Government Greenhouse Gas Reduction Project* (Ministry of Environment & Korea Environment Corporation, 2024), the average greenhouse gas emission intensity of passenger-car travel is 0.2111 $kgCO_2eq$ per passenger-kilometre. This value considers fuel-use shares, fuel-specific greenhouse gas emission factors, and energy consumption per passenger-kilometre. The calculation is as follows:

$[(72{,}305 \times 0.5378) + (75{,}391 \times 0.3736) + (64{,}464 \times 0.0884) + (133{,}000 \times 0.0003)] \times 2.90 \times 10^{-6} \approx 0.2111\ kgCO_2eq/passenger\text{-}km$

Fuel order: gasoline, diesel, LPG, and electricity. Emission factors are in $kgCO_2eq/TJ$; energy consumption is in TJ/passenger-km.

2. Scenario assumptions and calculation

To assess the potential for additional carbon emissions from reduced Ttareungyi usage, the scenario assumes that higher PM2.5 levels reduce cycling distance and that some of this distance is replaced by passenger-car travel. Ttareungyi travel distance is assumed to decrease by 1,000 km, with 10%, 30%, or 50% replaced by car travel. Cars are assumed to carry one person, and the replacement travel distance is assumed to equal the original cycling distance. Car emissions are estimated using the average per-passenger emission intensity above.

Additional greenhouse gas emissions ($kgCO_2eq$)

= Replacement car distance × 1 passenger × Car emission intensity

= Lost Ttareungyi distance (km) × Car replacement share × 1 passenger × 0.2111 ($kgCO_2eq$/passenger-km)

Car replacement share	Replacement distance	Additional emissions ($kgCO_2eq$)
10%	100 km	21.11

30%	300 km	63.33
50%	500 km	105.55

3. Interpretation and policy implications

The scenario analysis estimates additional greenhouse gas emissions of approximately 21.11, 63.33, and 105.55 kgCO_2eq if 10%, 30%, or 50% of the 1,000 km reduction in Ttareungyi travel is replaced by car travel, respectively. These are scenario results based on assumptions, rather than measured values. Higher PM2.5 levels do not necessarily imply increased car use, but policies should support public transport and environmentally friendly travel to minimize greenhouse gas emissions during periods of poor air quality. This approach suggests that public bicycle use can be linked to both air quality and long-term carbon emissions in the transport sector.

Reference

Ministry of Environment, & Korea Environment Corporation. (2024). 지자체 온실가스 감축사업별 감축원단위 적용 가이드라인 *[Guidelines for applying emission reduction units by local government greenhouse gas reduction project]*. https://www.kwaste.or.kr/data/06.php?admin_mode=read&no=5524&page=1